

AGENTIC AI PLATFORM FOR INTELLIGENT RESEARCH PAPER RETRIEVAL

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ABSTRACT

ResearchNexus: An Agentic AI Platform For Intelligent Research Paper Retrieval

Research papers are essential for academic learning and innovation, but searching for relevant papers and understanding lengthy technical documents can be time-consuming. ResearchNexus is an Agentic AI-powered platform designed to simplify the research process by enabling users to search research papers from trusted academic repositories such as arXiv, upload research paper PDFs, generate concise summaries, and interact with the uploaded documents through natural language questions using Retrieval - Augmented Generation (RAG) and Large Language Models (LLMs).

The platform also provides features such as research paper comparison, key point extraction, research gap identification, AI-generated presentation outlines, and citation generation, helping users understand research papers more efficiently and reduce the time required for literature review. The proposed system is developed using Python, Streamlit, LangChain, FAISS Vector Database, arXiv API, and a Large Language Model (LLM) to provide an intelligent and user-friendly research environment. ResearchNexus serves as a smart academic research assistant that enhances research productivity through AI-driven automation.